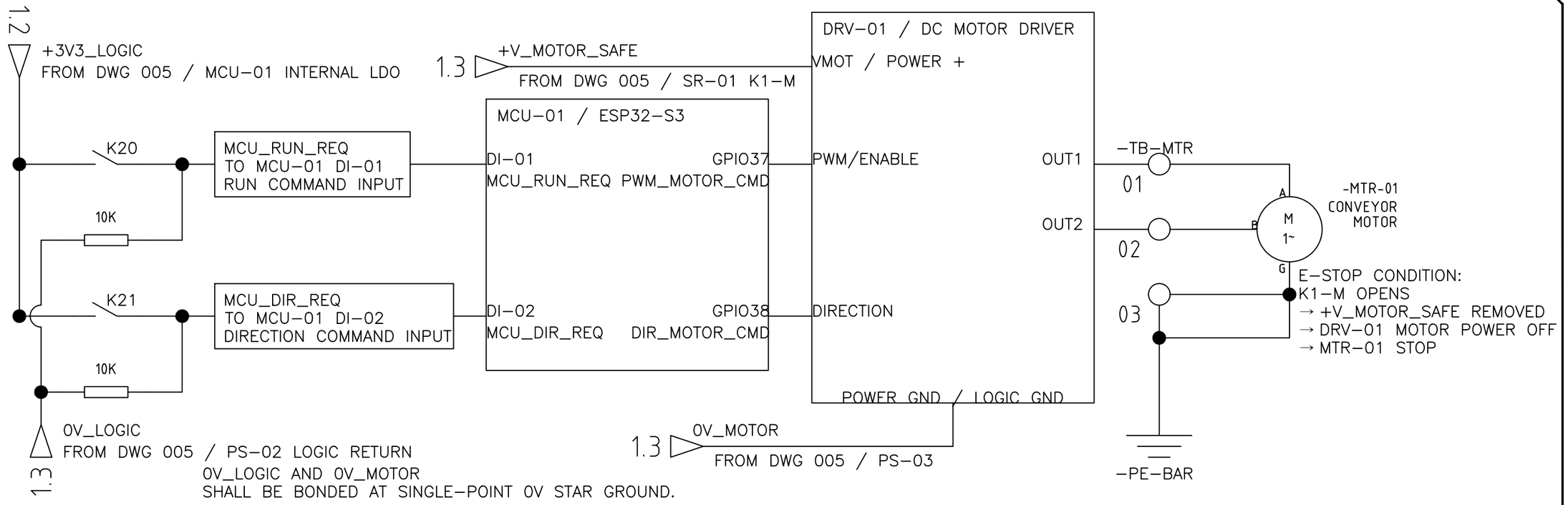


OUTPUT TERMINAL ALLOCATION

Terminal Strip	Terminal No.	Signal / Load	From	To	Supply / Return
TB-OUT-ACT	01	SOL-01 Clamp Command	K22 NO	SOL-01 Coil +	+24V_SAFE_ACT
TB-OUT-ACT	02	SOL-02 PASS Gate Command	K23 NO	SOL-02 Coil +	+24V_SAFE_ACT
TB-OUT-ACT	03	SOL-03 FAIL Gate Command	K24 NO	SOL-03 Coil +	+24V_SAFE_ACT
TB-OUT-ACT	04	SOL-04 RETEST Gate Command	K25 NO	SOL-04 Coil +	+24V_SAFE_ACT
TB-0V-ACT	01	SOL-01 Return	SOL-01 Coil -	0V Return Bus	0V_CTRL
TB-0V-ACT	02	SOL-02 Return	SOL-02 Coil -	0V Return Bus	0V_CTRL
TB-0V-ACT	03	SOL-03 Return	SOL-03 Coil -	0V Return Bus	0V_CTRL
TB-0V-ACT	04	SOL-04 Return	SOL-04 Coil -	0V Return Bus	0V_CTRL
TB-OUT-IND	01	LT-01 Green Command	K26 NO	Tower Lamp Green	+24V_CTRL_OUT
TB-OUT-IND	02	LT-01 Yellow Command	K27 NO	Tower Lamp Yellow	+24V_CTRL_OUT
TB-OUT-IND	03	LT-01 Red Command	K28 NO	Tower Lamp Red	+24V_CTRL_OUT
TB-OUT-IND	04	BZ-01 Buzzer Command	K29 NO	Buzzer +	+24V_CTRL_OUT
TB-0V-IND	01	Green Return	LT-01-G -	0V Return Bus	0V_CTRL
TB-0V-IND	02	Yellow Return	LT-01-Y -	0V Return Bus	0V_CTRL
TB-0V-IND	03	Red Return	LT-01-R -	0V Return Bus	0V_CTRL
TB-0V-IND	04	Buzzer Return	BZ-01 -	0V Return Bus	0V_CTRL
TB-MTR	01	Motor Output A	DRV-01 OUT1	MTR-01 A	Driver Output
TB-MTR	02	Motor Output B	DRV-01 OUT2	MTR-01 B	Driver Output
TB-MTR	03	Motor PE / Frame	PE-BAR	MTR-01 Frame	Protective Earth

[GENERAL DESIGN NOTE]

[1] PLC OUTPUTS P20 THROUGH P29 ENERGIZE INTERPOSING RELAY COILS K20 THROUGH K29.

[2] K20 AND K21 CONTACTS ARE DRY-CONTACT PLC COMMAND INTERFACES TO MCU-01.

[3] DO NOT APPLY 24VDC PLC OUTPUT VOLTAGE DIRECTLY TO MCU-01 / ESP32 GPIO INPUTS.

[4] MCU-01 SHALL GENERATE PWM AND DIRECTION SIGNALS FOR DRV-01 MOTOR DRIVER.

[5] DRV-01 SHALL BE FED FROM +V_MOTOR_SAFE. MOTOR POWER SHALL BE REMOVED BY K1-M DURING EMERGENCY STOP CONDITION.

[6] TB-MTR:01 AND TB-MTR:02 ARE MOTOR OUTPUT TERMINALS FROM DRV-01 TO MTR-01.

[7] TB-MTR:03 SHALL BE CONNECTED TO PE-BAR AND THE MTR-01 FRAME OR CABLE SHIELD.

[8] SOL-01 THROUGH SOL-04 SHALL BE FED FROM +24V_SAFE_ACT. ACTUATOR POWER SHALL BE REMOVED DURING E-STOP OR SAFETY CHAIN OPEN.

[9] LT-01 AND BZ-01 SHALL BE FED FROM +24V_CTRL_OUT TO ALLOW ALARM INDICATION DURING SAFETY ACTUATOR POWER REMOVAL.

[10] VERIFY ACTUAL MOTOR DRIVER INPUT VOLTAGE, MOTOR RATING, OUTPUT CURRENT, AND TERMINAL ASSIGNMENT BEFORE PANEL FABRICATION.

[CROSS REFERENCE]

K20 COIL: SEE DWG 008 / SHEET 1
K21 COIL: SEE DWG 008 / SHEET 1
PLC OUTPUT RELAY COIL CIRCUIT: SEE DWG 008 / SHEET 1

NO.	DATE	REVISION
A	2026-09-01	INITIAL RELEASE

DWG TITLE

EDS Wafer Map Linked Package
Final Test and Multi-Bin Sorting Cell

PLC Digital Output Circuit

DRAWN BY	DATE
J. H. KO	2026-09-01

DWG NO
008

SHEET NO
2 OF 2